

Linux

Linux being an open source OS gives you the power to alter the system to your needs. But this power must be utilised with caution. As the 'root' (Administrator) user of your machine you must take care that you do not corrupt the config files and scripts otherwise you may end up with a malfunctioning system.

Speeding up boot process

Most new users install a particular distro and use it with all the default process running. Processes such as pcmcia, nfs, sshd etc. are not required on a home desktop machine; these processes are mainly used on a server and some can be used on a desktop if you have a small network running a Squid proxy. The normal boot process of Linux will start most of these unwanted processes resulting in a long wait. It is easy to turn these unwanted processes off, especially on a Redhat-based system.

Use the command **#chkconfig -list** to display the list of process and its status for levels 0, 1, 2, 3, 4, 5 and 6. Of all these levels only 3 and 5 are concerned with desktop user operation. Instead of suggesting that you disabled these processes, we would rather tell you what processes are necessary for ideal operation. The essential processes are anacron, crond, echo, gpm, keytable, network (enable this if the PC belongs to a network), random, sendmail, sshd, syslog and xfs. You may enable other process that you think are necessary, say, for instance, kudzu is a hardware detection process that takes considerable time in the boot process; most of us don't open our machines and replace the PCI cards too often so kudzu can be disabled. To disable a process you will have to use 'chkconfig' command repeatedly for every process that needs a change in its status. The complete

command is **#chkconfig --level <levels 3 or 5> <process name> on|off**

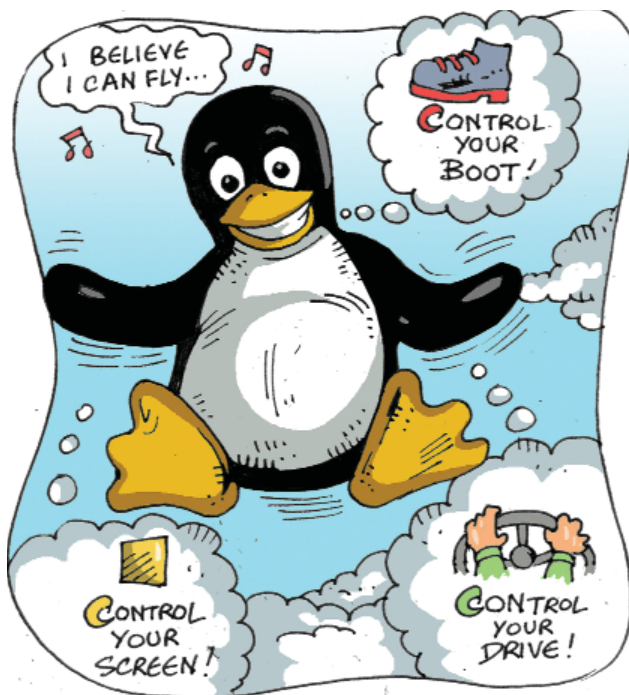
eg: **# chkconfig --level 5 kudzu off**

For Redhat users this task is extremely simple; thanks the nifty use of ncurses. Typing the command 'ntsysv' on a konsole prompt will list out all the process with its status indicated in '['*']' box preceding the process name; use spacebar button to remove the '*' in the box, this

will disable the respective process.

Some of you may wonder what a level is. Level 3 boots the machine in text mode while level 5 is the GUI mode that directly starts X-server and the X-windows environment (KDE or GNOME).

Once the necessary processes are set, reboot the system, you will see a marked difference in the boot up time.



Speeding up IDE-based drives

This tip is useful for just about any Linux desktop machine using IDE-based drives. By default Linux uses conservative settings for IDE for instance the DMA mode is disabled. This is done to achieve stability at the cost of speed. But there is no cause for concern, as you can change the settings through a command line utility called hdparm. Do **#man hdparm** for details regarding this feature.

Out of the many options it suggests we will look at only a few that will help us improve performance of the system. Before we start making changes to a drive's setting we should look at its default setting. If you are logged in as a user, then logout and log back in as root. Now, type **#hdparm -v /dev/hda** and hit [Enter]. This command will display the current device setting of your hard drive. For those who are not aware: hda = primary master, hdb = primary slave, hdc = secondary master and hdd = secondary slave. On our system running Redhat (Shrike) on Seagate 40GB HDD (IDE) we got the following output:

```
[root@aryan rc.d]#  
hdparm -v /dev/hda  
/dev/hda:  
multcount   = 16 (on)  
IO_support  = 0 (default  
16-bit)  
unmaskirq   = 0 (off)  
using_dma   = 0 (off)  
keepsettings = 0 (off)  
readonly    = 0 (off)  
readahead   = 8 (on)  
geometry    =  
4865/255/63, sectors =  
78163247, start = 0
```

Most of the new hard drives will display 'using_dma = 1' which is good and that's exactly what we will do here.

To enable DMA mode on a hard drive use the flag '-d 1' command along with the command hdparm. Similarly, we will add flags viz. -a, -A, -d, -X, -m and -u. Before we continue, let us warn you that manipulating drive settings can cause problems if your chipset is not supported